Development of a Data-Assimilative Digital Twin for Wave Energy Converters Using Field, Laboratory and Numerical Data

Abstract

This project aims to develop a data-assimilative digital twin framework for wave energy converter (WEC) operation by integrating physics-based hydrodynamic models with PacWave field measurements, experimental data from multiple WEC devices (subWEC, LUPA, FOSSWEC) tested at the Hinsdale Wave Research Laboratory, and PTO control-system data from the WESURF laboratory. Leveraging existing computational workflows, calibrate and validate numerical models and the requested computing resources, the project will assimilate measurements to update model states and parameters and quantify predictive uncertainty. Key outputs will include WEC motion and PTO response estimates, power-production predictions and model-performance diagnostics. The framework will establish a computational foundation for improved WEC performance assessment, predictive operation and future control optimization under realistic ocean conditions.

1 Project Overview and Research Opportunity

This project proposes to develop a data-assimilative digital twin for wave energy converters (WECs) that connect ocean wave measurements, laboratory experiments and numerical simulations to predict device response and power conversion across scales. The framework shown in Fig. 1 will combine full-scale wave data from PacWave, scaled WEC hydrodynamic experiments at Oregon State University's O.H. Hinsdale Wave Research Laboratory and power take-off (PTO) bench experiments at WESURF to create a comprehensive model of WEC behavior. Measured wave conditions from PacWave will be scaled to laboratory conditions and used as inputs to physics-based WEC models to predict device motions and hydrodynamic forces. Measurements from Hinsdale wave-tank experiments and WESURF PTO tests will then be combined with the numerical predictions through data assimilation to improve estimates of device response, calibrate selected model parameters and assess prediction uncertainty. Building on existing numerical tools and WESURF's force/position driven and coupled testing capabilities, the project will establish and validate a generalized, reusable workflow linking tank testing, hydrodynamic simulation with physical PTO testing. The resulting framework will support assessment of loads on devices, dynamics and power output under representative ocean conditions. This workflow will provide a foundation for future integration of operational measurements from deployed devices.

2 Previous Research and Integration Gap

Digital twins are virtual representations of physical systems that can be used to simulate and optimize their behavior. The development of digital twins for wave energy converters (WECs) has

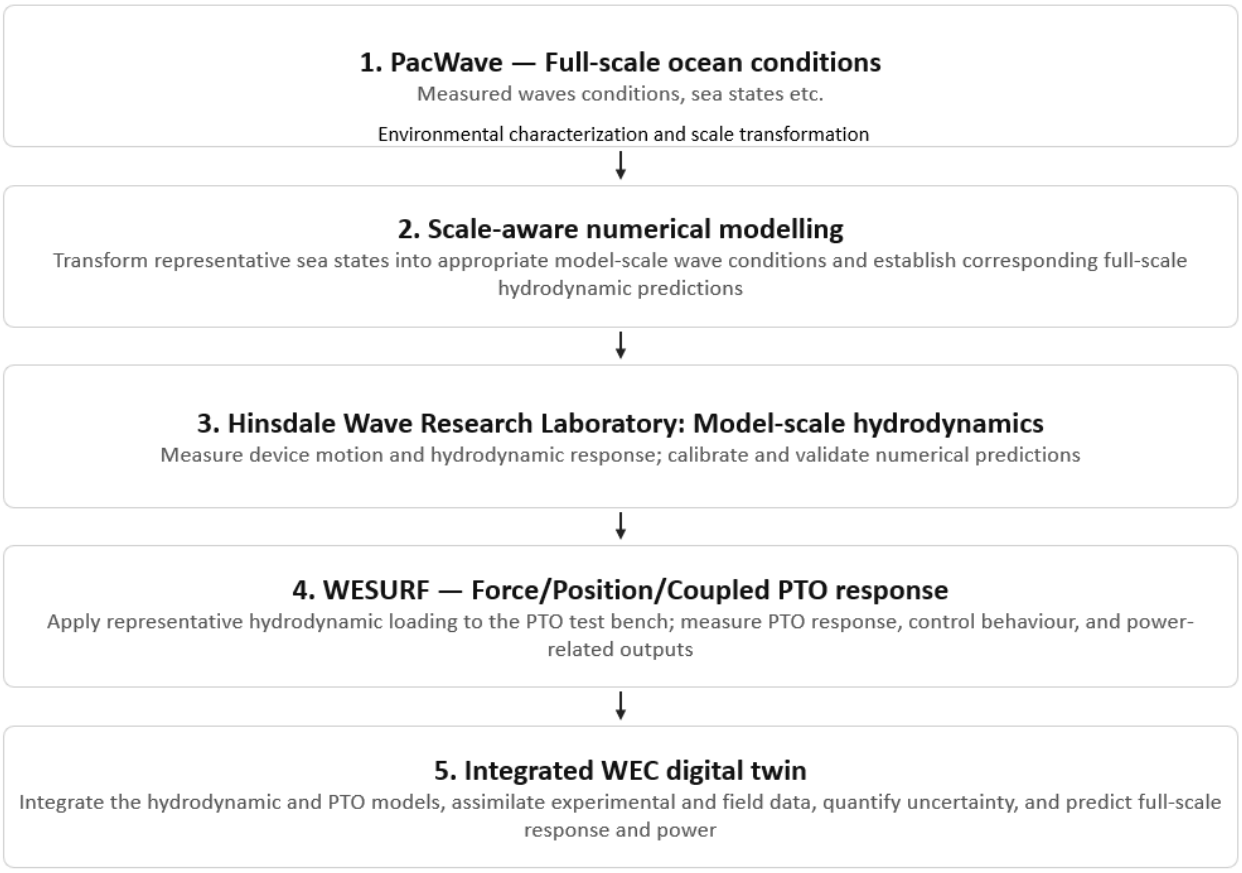


Figure 1: Framework for the proposed data-assimilative digital twin for wave energy converters.

gained significant attention in recent years due to the increasing need for accurate and reliable prediction of device performance under real-world conditions. A digital twin can help in understanding the complex interactions between the ocean environment, the WEC itself, and its control systems. This understanding is crucial for improving the design, operation and maintenance of WECs.

In a digital twin framework experimental data is used to calibrate and validate numerical models, which can then be used to predict the behavior of the physical system under different conditions. PacWave PacWave (2026) wave resource data was characterized by Dunkle et al. (2020) and Robertson et al. (2022) examined how spectral conditions at PacWave and other sites influence device-size-dependent energy production and variability. At field scale, Okushemiya et al. (2026) combined deployment measurements with PTO dynamometer data to improve modelling of a tethered-heave-plate WEC, incorporating generator inertia and measured power-conversion behaviour. Ali et al. (2025, 2026) in several tank-testing campaigns at OSU investigated the behavior of submerged-WEC's under various conditions. Nonlinear surge–pitch coupling and subharmonic excitation was investigated respectively in these studies, highlighting the importance of considering nonlinear effects in WEC modeling. McConnell et al. (2025, 2026) developed and experimentally evaluated real-time hybrid approaches for WEC mooring dynamics using Hinsdale Wave Research Laboratory and PTO bench experiments at WESURF. Additionally, the OSU–Sandia–Evergreen FOSTWIN Sandia National Laboratories et al. (2026) platform offers both a physics-based WEC-Sim model and a model identified from Hinsdale experimental data for controller development. Collectively, these studies provide the data and methodologies to motivate the proposed integration of

PacWAVE field wave conditions, tank-validated hydrodynamics and physical PTO testing within a reusable data-assimilation workflow.

Outside the Pacific Marine Energy Center (PMEC), similar efforts are being made to develop digital twins for wave energy converters. At ocean scale, Dizon et al. (2024) evaluated WEC-Sim and Gazebo models of the deployed MBARI two-body WEC against field measurements across 56 observations, comparing piston velocity, motor speed, and electrical power; the Gazebo model was intended to serve as a digital twin. Previous research provides complementary foundations for WEC digital twins through experimental model identification, field validation, and coupled PTO testing. Giorgi et al. (2019) used six experimental datasets to identify and validate efficient linear and nonlinear hydrodynamic models, demonstrating how laboratory measurements can inform predictive models. Lawal et al. (2026) discuss AI-driven digital twins for wave energy systems, while Katsidoniotaki et al. (2022) developed a CFD-derived surrogate for rapid prediction of extreme mooring loads. Related computational approaches include nonlinear response prediction using multiple-input operator networks by Zhang et al. (2025), CFD-coupled reinforcement learning for adaptive PTO control by Liang et al. (2024), and data-driven control methods reviewed by Pasta et al. (2023). For physical PTO testing, Bracco et al. (2015) coupled an ISWEC gyroscope and generator test rig with a numerical hydrodynamic model through hardware-in-the-loop simulation, while Bacelli et al. (2019) demonstrated control testing through complementary dry-bench and wave-tank experiments. Together, these studies motivate a framework connecting measured ocean-wave conditions, tank-validated hydrodynamics, and physical PTO response.

The proposed project will build on these approaches by developing a reusable workflow that explicitly assimilates laboratory measurements to update model estimates and assess prediction uncertainty across the hydrodynamic and PTO testing stages.

3 Research Objectives and Expected Outcomes

The project has five linked objectives.

1. Identify the representative wave conditions derived from PacWave measurements.
2. Organize the existing tank-testing and PTO (Pneumatic) datasets into a documented structure that preserves device geometry, scale, instrumentation, wave conditions and PTO settings.
3. Identify the most appropriate numerical models (WEC-Sim, CFD) that reproduce selected laboratory experiments and validate their predictions against independent datasets.
4. Develop and evaluate a data-assimilation method that uses measured device response to update estimated states and selected model parameters.
5. Integrate the resulting hydrodynamic model with WESURF's physical PTO testing capabilities and evaluate the coupled response.

Expected outcomes include a curated dataset inventory, validated baseline and updated models, a documented coupled simulation and testing workflow and an assessment of prediction accuracy and uncertainty. Success will be demonstrated through reproducible comparisons of device motion, PTO displacement or velocity, measured connection loads and power metrics.

4 Research Approach: Connecting Field, Tank, and PTO Data

5 Computational Readiness and Resource Requirements

6 Work Plan, Milestones, and Validation

7 Impact Readiness and Pathway to External Funding

References

Raza Ali, Aeron Roach, Bret Bosma, and Bryson Robertson. Experimental analysis of the effect of surge-pitch coupling on the dynamics and power production a submerged wave energy converter. In *Proceedings of the European Wave and Tidal Energy Conference*, volume 16, 2025. doi:10.36688/ewtec-2025-881. URL `https://submissions.ewtec.org/proc-ewtec/article/view/881`.

Raza Ali, Aeron Roach, Bret Bosma, Pedro Lomonaco, and Bryson Robertson. Subharmonic excitation and nonlinear coupling in submerged wave energy converters. *Ocean Engineering*, 368:128277, 2026. ISSN 0029-8018. doi:https://doi.org/10.1016/j.oceaneng.2026.128277. URL `https://www.sciencedirect.com/science/article/pii/S0029801826041119`.

Giorgio Bacelli, Steven J. Spencer, David C. Patterson, and Ryan G. Coe. Wave tank and benchtop control testing of a wave energy converter. *Applied Ocean Research*, 86:351–366, 2019. doi:10.1016/j.apor.2018.09.009.

Giovanni Bracco, Ermanno Giorcelli, Giuliana Mattiazzo, Vincenzo Orlando, and Mattia Raffero. Hardware-in-the-loop test rig for the ISWEC wave energy system. *Mechatronics*, 25:11–17, 2015. doi:10.1016/j.mechatronics.2014.10.007.

Chris Dizon, Ryan Coe, Andrew Hamilton, Dominic Forbush, Michael Anderson, Ted Brekken, and Giorgio Bacelli. Analysis on evaluations of monterey bay aquarium research institute's wave energy converter's field data using WEC-Sim and Gazebo: A simulation tool comparison. *Applied Sciences*, 14(23):11169, 2024. doi:10.3390/app142311169.

Gabrielle Dunkle, Bryson Robertson, Gabriel García-Medina, and Zhaoqing Yang. PacWave wave resource assessment. Technical report, Pacific Marine Energy Center, Oregon State University, 2020. URL `https://pacwaveenergy.org/wp-content/uploads/2021/11/PacWaveRA_updated.pdf`. Updated November 2020.

Simone Giorgi, Josh Davidson, Morten Jakobsen, Morten Kramer, and John V. Ringwood. Identification of dynamic models for a wave energy converter from experimental data. *Ocean Engineering*, 183:426–436, July 2019. ISSN 00298018. doi:10.1016/j.oceaneng.2019.05.008. URL `https://linkinghub.elsevier.com/retrieve/pii/S0029801818304748`.

Eirini Katsidoniotaki, Foivos Psarommatis, and Malin Göteman. Digital Twin for the Prediction of Extreme Loads on a Wave Energy Conversion System. *Energies*, 15(15):5464, July 2022. ISSN 1996-1073. doi:10.3390/en15155464. URL `https://www.mdpi.com/1996-1073/15/15/5464`.

Zaharaddeen Karami Lawal, Hayati Yassin, Rufai Yusuf Zakari, and Azam Che Idris. AI-driven digital twin for sustainable management of wave energy conversion systems. In *Dig-*

ital Twins for Sustainable Development, pages 103–128. Elsevier, 2026. ISBN 978-0-443-27388-9. doi:10.1016/B978-0-443-27388-9.00012-X. URL https://linkinghub.elsevier.com/retrieve/pii/B978044327388900012X.

Hongjian Liang, Hao Qin, Haowen Su, Zhixuan Wen, and Lin Mu. Environmental-Sensing and adaptive optimization of wave energy converter based on deep reinforcement learning and computational fluid dynamics. *Energy*, 297:131254, June 2024. ISSN 03605442. doi:10.1016/j.energy.2024.131254. URL https://linkinghub.elsevier.com/retrieve/pii/S0360544224010272.

Abilyn McConnell, Bryson Robertson, and Bret Bosma. Experimental validation of a real-time hybrid mooring system for wave energy systems using a linear test bed. In *Proceedings of the European Wave and Tidal Energy Conference*, volume 16, 2025. doi:10.36688/ewtec-2025-703. URL https://submissions.ewtec.org/proc-ewtec/article/view/703.

Abilyn McConnell, Elaine Liu, Bret Bosma, Inyong Kim, Raza Ali, Barbara Simpson, and Bryson Robertson. Development of a real-time hybrid simulation framework for wave energy converter mooring applications. *Ocean Engineering*, 362:126498, 2026. doi:10.1016/j.oceaneng.2026.126498.

PacWave. Pacwave: Open-water wave energy testing. https://pacwaveenergy.org/, 2026. Accessed 2026-10-13.

Edoardo Pasta, Nicolás Faedo, Giuliana Mattiazzo, and John V. Ringwood. Towards data-driven and data-based control of wave energy systems: Classification, overview, and critical assessment. *Renewable and Sustainable Energy Reviews*, 188:113877, December 2023. ISSN 13640321. doi:10.1016/j.rser.2023.113877. URL https://linkinghub.elsevier.com/retrieve/pii/S1364032123007359.

Bryson Robertson, Gabrielle Dunkle, Tim Mundon, and Levi Kilcher. Wave resource spatial and temporal variability dependence on WEC size. *International Marine Energy Journal*, 5(1): 113–121, 2022. doi:10.36688/imej.5.113-121. URL https://marineenergyjournal.org/imej/article/view/104.

Sandia National Laboratories, Oregon State University, and Evergreen Innovations. FOSTWIN: Digital twin of the floating oscillating surge wave energy converter. Software repository, 2026. URL https://github.com/PMEC-OSU/FOSTWIN. Accessed October 13, 2026.

Qi Zhang, Haicheng Zhang, Xin Zhao, Jun Ding, and Daolin Xu. Multiple-input operator network prediction method for nonlinear wave energy converter. *Ocean Engineering*, 317: 120106, February 2025. ISSN 00298018. doi:10.1016/j.oceaneng.2024.120106. URL https://linkinghub.elsevier.com/retrieve/pii/S0029801824034449.